

Features of ECBC

- ECBC covers energy usage per unit of floor space and other high energy consuming equipments like HVAC, lighting
- ECBC details the parameters of various building material to be used to achieve overall performance of building envelope so as to minimise heat gain and thus the cooling cost
- ECBC specifies use of energy efficient glass combinations to maximise daylight and minimise cooling loads

ECBC provisions are designed to enhance comfort and occupant productivity. ECBC encourages energy efficient design or retrofit of commercial buildings so that they are designed in a manner that reduces the use of energy without affecting the building function, the comfort, health, or the productivity of the occupants. These codes eliminate building design practices that lead to unnecessarily high building energy use and associated costs. Energy cost savings resulting from energy code compliance directly benefit building owners and occupants over the life cycle of the building.

The scope of ECBC covers energy usage per unit of floor space and other high energy consuming equipments like HVAC, lighting. It also details the parameters of various building material to be used to achieve overall performance of building envelope so as to minimise heat gain and thus the cooling cost. ECBC specifies use of energy efficient glass combinations to maximise daylight and minimise cooling loads. However, the broad requirements of the Code with respect to the building envelope are the same for new buildings as well as for extensions and modifications

ECBC provisions are designed to enhance comfort and occupant productivity. ECBC encourage energy efficient design or retrofit of commercial buildings so that they are designed in a manner that reduces the use of energy without affecting the building function, the comfort, health, or the productivity of the occupants and with appropriate regard for economic considerations. These codes eliminate building design practices that lead to unnecessarily high building energy use and associated costs. Energy cost savings resulting from energy code compliance directly benefit building owners and occupants over the life cycle of the building.

All over the world ECBCs have a proven track record of significantly reducing energy use in buildings in a highly cost effective way. The ECBC benefit both individual building owners and the people and government at large. Benefits to property owners include reduced energy costs and improved comfort (both thermal and visual).

The benefits to society include the following:

- Reduced capital investments in energy supply infrastructure
- Reduced environmental impacts

- Improved electricity reliability
- More efficient use of resources (It is cheaper to save energy than it is to build new power plants)

It is estimated that buildings that are ECBC complaint have a potential of savings to the tune of about 30%-35%

ECBC- Benefits

In India, estimates suggest that about 20 to 25% of the total energy demand is due to manufacturing materials required in the construction sector, while another 15% goes into the running needs of the building. The proposed Energy Conservation Building Code sets a minimum efficiency standard for commercial buildings in all climate zones in the country. The estimated reduction in energy use for new buildings range between 25% to 40% depending on climate, building type, and hours of operation.

ECBC benefits both the individual building owners and the society. Benefits to property owners include reduced energy costs and improved comfort (both thermal and visual). The benefits to society include reduced capital investments in energy supply infrastructure, reduced environmental impacts, improved electricity reliability, and more efficient use of resources.

The Government will take necessary measures to ensure proper dissemination of information about ECBC compliance procedures while making it consumer friendly. It also commits to identify appropriate methods of encouraging compliance, including monetary awards, publicity awards, building energy labels, and technical assistance.

Road Map to Make ECBC Mandatory

Given the fact that the capacity in the country required to effectively implement this code is inadequate, the implementation of the codes will be on voluntary basis initially. Incentives to promote its use in the voluntary phase will be provided alongwith promoting the industry for insulators, windows, etc. Only when there is sufficient availability of both technical expertise and complaint material, will the codes be made mandatory. The Government will launch an effective awareness campaign to promote ECBC all over the country.

Energy Conservation Building Codes (ECBC) are delivering significant and lasting energy savings in new construction where they have been implemented with care. ECBC are rightly considered as a pillar to any government energy efficiency and climate change policy. ECBC benefits both the individual building owners and the society. Benefits to property owners include reduced energy costs and improved comfort (both thermal and visual). The benefits to society include reduced capital investments in energy supply infrastructure, reduced environmental impacts, improved electricity reliability, and more efficient use of resources.

ENERGY IS LIFE

